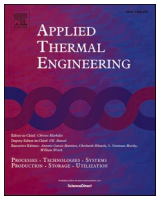
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Research Paper



Research on the thermal performance and stability of three-dimensional array pulsating heat pipe for active/passive coupled thermal management application

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ABSTRACT

The efficient heat transfer performance of three-dimensional pulsating heat pipes can effectively deal with the heat dissipation problems in the field of electronics integration. However, the large contact thermal resistance and complex layout limit the actual cooling efficiency. In this study, a novel three-dimensional array pulsating heat pipe (3D-APHP) with a dual-plane structure was designed. The top and bottom of the 3D-APHP are mounted with flat plates, and the center section is connected by multiple parallel array tubes. The planar and array structures enhance the flexibility in the actual layout of the pulsating heat pipe to accommodate space constraints. The effects of different conditions (working fluid, filling rate, heating power, tube diameter and installation condition) on the heat transfer performance and uniform temperature performance of 3D-APHP were investigated. It is found that 3D-APHP demonstrates efficient heat transfer capability and temperature uniformity. The minimum thermal resistance and maximum thermal conductivity can reach 0.059 °C/W and 170.1 KW/m °C, respectively. Moreover, 3D-APHP exhibits good heat transfer performance across different installation positions with a maximum thermal resistance not exceeding 1.1 °C/W, revealing significant potential for applications. Meanwhile, the two-phase heat transfer process inside the 3D-APHP was analyzed. The uniform vapor–liquid plug distribution can significantly strengthen the thermal performance of the 3D-APHP.

1. Introduction

With the continuous development of technology in the pillar industries of energy, important technologies such as pneumatic tubes, space shuttles, and 5G base stations are becoming increasingly sophisticated. The increasing prevalence of electronic devices and equipment with high levels of integration and power consumption has become ubiquitous [1]. The increased integration of electronic devices will directly lead to limited space for heat dissipation, resulting in concentrated heat flow density, low heat dissipation efficiency and poor operational reliability. There is an urgent need for thermal management systems that can effectively address heat dissipation issues in small and efficient heat transfer devices [2,3].

As key components with high thermal conductivity performance, heat pipes (HPs) play an important role in the cooling systems of electronic devices due to their stable operational performance. Among the various HP technologies, the sintered heat pipe (SHP) is a widely adopted form [4]. The internal structure of the SHP, primarily made of high thermal conductivity materials powdered and sintered to create a porous framework, enhances its heat transfer capacity by facilitating the pumping of condensed fluid. Li et al [5] developed a SHP with dual parallel condensers for cooling high-power integrated LED chips. The test showed that the junction temperature of the LED was controlled below 85 °C under a thermal load of 300 W. Okutani et al [6] designed a multi-evaporator multi-condenser HP with polytetrafluoroethylene porous media core. The SHP showed stable start-up and efficient operation. Cai [7] engineered a multi-branch SHP design. Experimental data indicated that the peak temperature of the system could be sustained below 115 °C with this SHP. Zhong et al. [8] and Khalili and Shafii [9], who both investigated the thermal performance of the new SHP with multi-branch. They found that the newly designed SHP has the capacity to reduce thermal resistance. Zhao et al. [10] designed a new sintered multi-size copper powder core, and experimentally found that the overall thermal resistance of the HP was reduced by 14.22 %. Wang et al. [11] designed a SHP with sintered copper particles added inside the evaporator. The new SHP sped up the cooling process for high-power LED chip heat dissipation. SHP effectively mitigates heat flux

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Nomenclature

D	Diameter [mm]
Di_{max}	Critical pipe diameter [mm]
L	Length [mm]
Q	Heating power [W]
R	Thermal resistance [°C /W]
T	Temperature [°C]
η	Uniform temperature coefficient
λ	Thermal conductivity [W/(m•°C)]

Subscripts

ave	Average
a	Adiabatic section
c	Condensation
e	Equivalent or evaporation
i	inner
liq	liquid
max	Maximum
min	Minimum
vap	vapor

Greek Symbol

g	gravity [N]
ρ	density [kg/m^3]
σ	surface tension [mN/m]

Acronyms

FR	Filling rate
HP	Heat pipe
PHP	Pulsating heat pipe
SHP	Sintered heat pipe
3D-PHP	Three-dimensional pulsating heat pipe
3D-APHP	Three-dimensional array pulsating heat pipe

concentration. However, the thermal conductivity of SHP is limited by its internal structure and materials, leading to a restricted application temperature range and higher costs.

Pulsating heat pipe (PHP) is a new member of the HP family. Compared to conventional HPs, PHPs are not affected by the entrapment limit due to their simplified structure devoid of capillary components [12–14], enhancing their heat transfer efficacy. Through leveraging both latent and sensible heat via "self-excited" oscillatory motion of the working fluid, PHPs ensure effective thermal management in electronic devices [15]. According to the strong structural adaptability and flexible arrangement of cold and heat sources, new structure of PHPs can be designed for different application requirements. As respect to the research of two-dimensional PHP: Mehta et al. [16] compared the heat transfer performance of PHP with rectangular and circular cross-sections at different sizes. The rectangular channel enhanced the oscillatory motion of the working fluid and reduced the thermal resistance of the PHP. Hao [17] and Chen et al. [18] similarly demonstrated that the sharp-angled structure of the rectangular cross-section can promote the pulsation of the working fluid within the PHP. Xiahou et al. [19] and Kong et al. [20] divided the PHP into upper and lower channels by spacers to form double-sided rectangular and triangular flat PHPs. The equivalent thermal conductivity of the new PHP was more than twice that of the conventional PHP, and the double-sided triangular PHP had better heat transfer performance. Jung et al. [21] developed a high-performance metal-sealed flexible PHP. The maximum effective thermal conductivity in the vertical direction is 1070 W/m·K, and the rate of increase of pressure inside the PHP is only 0.54 Pa/d with a lifetime of at least 2000 days. Abela et al. [22] designed a novel PHP made up of four borosilicate transparent glass tubes. It was found that increasing the number of evaporator sections and elevating the condenser temperature can effectively enhance the equivalent thermal resistance performance of the PHP. Patel et al. [23] conducted experimental research on the thermal performance of a non-symmetric single-turn closed PHP. The asymmetric adiabatic section could enhance the heat carrying capacity of water and acetone, the working fluids, by 75 % and 50 % respectively.

Facing the heat dissipation conditions with high uniform temperature demand, high heat flux, and complex spatial structure, the traditional PHP can hardly meet the requirements. The three-dimensional PHP (3D-PHP) emerges for practical application conditions [24]. Qu et al. [25,26] bent two-dimensional PHP into multilayer 3D-PHP. 4-layer PHP had better start-up and heat transfer performance than other layers of PHP. Lan [27] developed an innovative tubular 3D-PHP by constructing a closely spaced annular parallel channel using copper wire, which effectively improved the heat transfer performance of the PHP. The device attained its optimal thermal resistance of 0.131 °C/W when filled to 55 %. He et al. [28] investigated the performance and flow pattern of the annular 3D-PHP through visualization experiments. With the increase of thermal load, the start-up time of the PHP shortened and the thermal resistance decreased. Jang et al. [29] improved the thermal performance of the radial divergent channel 3D-PHP by adding a Tesla valve. The preferential motion in one direction increased the flow circulation, reducing the thermal resistance of the PHP by 17 %. Thompson et al. [30,31] explored a plate 3D-PHP through visualization and heat transfer studies. The initiation of PHP with the staggered distribution of channels on both sides of the plate was accompanied by temperature field relaxation and thermal oscillation. Three-dimensional compact design broadens the range of applications. Chen et al. [32] developed a flower-shaped PHP with a petal-like evaporator and a cylindrical condenser at its base. Experimental observations revealed that the flower-shaped PHP exhibited excellent operational performance, achieving a minimal thermal resistance of up to 0.14 °C/W. Pagliarini et al. [33] conducted experiments on a multi-loop 3D-PHP. They found that the investigated three-dimensional layout was gravity-independent under high power input, with the thermal resistance in almost every direction around 0.25 K/W. Alqahtani et al. [34] evaluated the impact of bending angles and the arrangement of evaporators and condensers on the heat transfer performance of a flexible polypropylene 3D-PHP. The arrangement did not significantly affect thermal performance, but it might influence the start-up of the PHP.

In the field of active/passive coupled thermal management systems based on PHPs, the conventional tubular 3D-PHP and flat-plate 3D-PHP have significant requirements for space and installation, and they suffer from uneven heat dissipation [35]. There are no existing reports that a 3D-PHP has completely solved these issues. This study proposes a novel three-dimensional array pulsating heat pipe (3D-APHP) with a dual-plane structure for the thermal management demand of high heat flow planes. The evaporation section and condensation section are designed as flat plates with a cavity inside and connected through several parallel array tubes. The utilization of a dual-plane structure conforms more closely to the flat heat source in practical applications, which reduces the contact thermal resistance and enhances the flexibility in arranging the hot and cold source. Additionally, the adoption of an array structure facilitates changing the tube length according to needs, thereby enabling long distance heat transfer and greatly enhancing temperature uniformity. The study systematically investigates the impact of working fluid type, filling rate (FR), heating power, tube diameter, and installation conditions on the start-up characteristics, heat transfer performance, and temperature uniformity of the 3D-APHP. Additionally, the internal two-phase flow heat transfer mechanism of the 3D-APHP is analyzed.

2. Experimental analysis

2.1. Experiment setup

Fig. 1 shows the schematic structure of the designed 3D-APHP and illustrates its advantages. The dual-plane structure and array structure can improve the temperature uniformity and enhance the flexibility in arranging the PHP within the system.

The main body of the 3D-APHP is composed of 12 straight copper tubes, 2 concave plates and 2 convex plates. The length of straight copper tubes is 20 cm, and the inner diameters were designed to be 3 mm and 4 mm respectively. The overall dimensions of the concave convex plates both are 17 × 7 × 0.5 mm^3 with 14 holes on the surface. The function is to connect the concave plate and the convex plate with fasteners. In addition, there is a cavity of size 15 × 5 × 0.3 mm^3 engraved on the surface of the concave plate to increase the volume of the working fluid. A liquid cooling chamber was installed at the top of the 3D-APHP for circulating water.

The physical schematic of the designed 3D-APHP is shown in Fig. 2 (a). The 3D-APHP was equipped with a pressure sensor and a two-way valve for fluid injection and vacuum evacuation. Fig. 2(b) shows a high-power insulated gate bipolar transistor module, with its internal heat dissipation being a primary research focus. The designed 3D-APHP could be installed at the bottom of the module as a heat dissipation element and filled with phase change material inside. It can ensure the uniform heat dissipation effect inside the module and maximize the solution to the heat accumulation problem.

Fig. 2(c) shows the experimental setup of the 3D-APHP test system, which mainly includes 3D-APHP, DC power supply, thermostatic water bath, pump, data acquisition system and PC. The electric heating element was installed at the bottom of the 3D-APHP with heat-conductive silicone grease in the middle. Twenty thermocouples were arranged on the 3D-APHP. The PHP was wrapped with insulation cotton and fixed with aluminum foil tape. The physical schematic of the system is shown in Fig. 2(d). The experiment investigated the heat transfer performance of 3D-APHP in vertical and horizontal orientations to affirm whether the 3D-APHP can efficiently dissipate heat across diverse installation conditions, thereby broadening its potential applications. Fig. 2(e) illustrates the different placement conditions of 3D-APHP. In addition, a visualized 3D-APHP was created with glass tubes replacing copper ones in the adiabatic section for detailed flow analysis using a high-speed camera, as depicted in Fig. 2(f).

Deionized water, anhydrous ethanol and Novec7000 are chosen as the working fluids in the experiment. Table 1 displays the thermal properties of the working fluids at room temperature and atmospheric pressure [36]. Among them, the critical pipe diameter (Di_{max}) significantly affects PHP heat transfer efficiency. A reasonable range of Di_{max} can allow PHP to adapt to different installation angles, and the internal vapor–liquid plugs are distributed at random and stable intervals. The theoretical Di_{max} calculation is shown in formula (1). The theoretical Di_{max} corresponding to the three working fluids has been listed in Table 1.

$$Di_{max} = 2\sqrt{\sigma/(\rho_{liq} - \rho_{vap})g} \tag{1}$$

2.2. Experimental procedure

FR should be considered before the experiment. The lower FR leads to the drying out of the evaporation section, while the higher FR increases the pressure drop. Previous studies reported that the PHP yields superior thermal performance at 20 % to 50 % FR [37,38]. However, the optimal FR may vary for different PHP structures. To explore this, nine FR groups ranging from 5 % to 70 % in 10 % intervals were tested under room temperature conditions (20 °C to 24 °C). Moreover, the experiment was conducted to remove non-condensable gases through repeated heating and freezing processes to ensure cleanliness inside 3D-APHP [39].

The experimental heating power ranged from 20 W to 100 W. In order to minimize heat loss in the system, the exterior of the 3D-APHP was insulated with foam and aluminum foil tape, reducing heat loss to less than 5 %. The heat dissipation of the 3D-APHP was taken away by cooling water provided by a thermostatic water bath at 15 °C. Twenty high-precision K-type thermocouples were pasted on the surface of the 3D-APHP to collect temperature data. To study the internal flow of the 3D-APHP, a high-speed infrared camera monitored the surface temperature distribution throughout operation. Time series analysis of these changes allowed for deducing the fluid flow state and evaluating the performance of the 3D-APHP. The conditions of each experimental test are summarized in Table 2.

2.3. Data and uncertainty analysis

The heat transfer performance of PHP is mainly evaluated by the key metrics of thermal resistance (R), equivalent thermal conductivity (λ_e), and start-up temperature, etc. The average temperature difference of the PHP can be calculated by formulas (2)~(4). The R can be calculated by formula (5). $T_1 \sim T_6$ are the temperatures measurement points of the evaporation section, with T_e as their average. $T_{13} \sim T_{18}$ are the temperature measurement points of the condensation section, with T_c as their average. T_{ave} represents the average temperature difference. Q is the input heating power.

$$\overline{T_e} = \frac{1}{6}(T_1 + T_2 + T_3 + T_4 + T_5 + T_6) \tag{2}$$

$$\overline{T_c} = \frac{1}{6}(T_{13} + T_{14} + T_{15} + T_{16} + T_{17} + T_{18}) \tag{3}$$

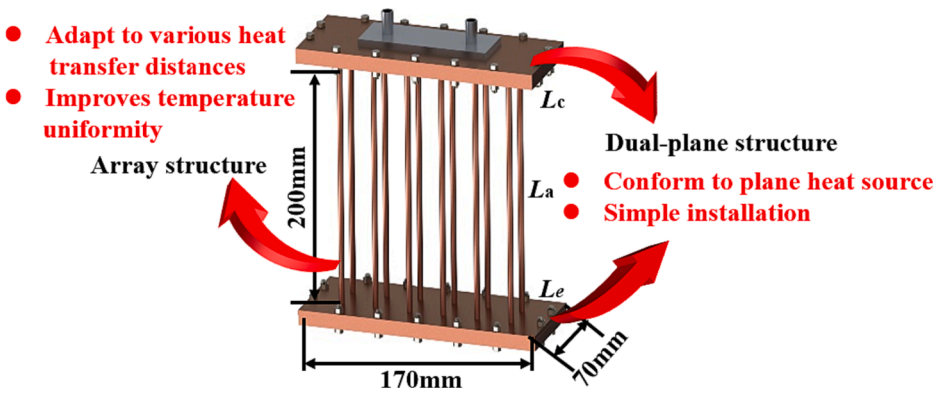


Fig. 1. Schematic structure of 3D-APHP.

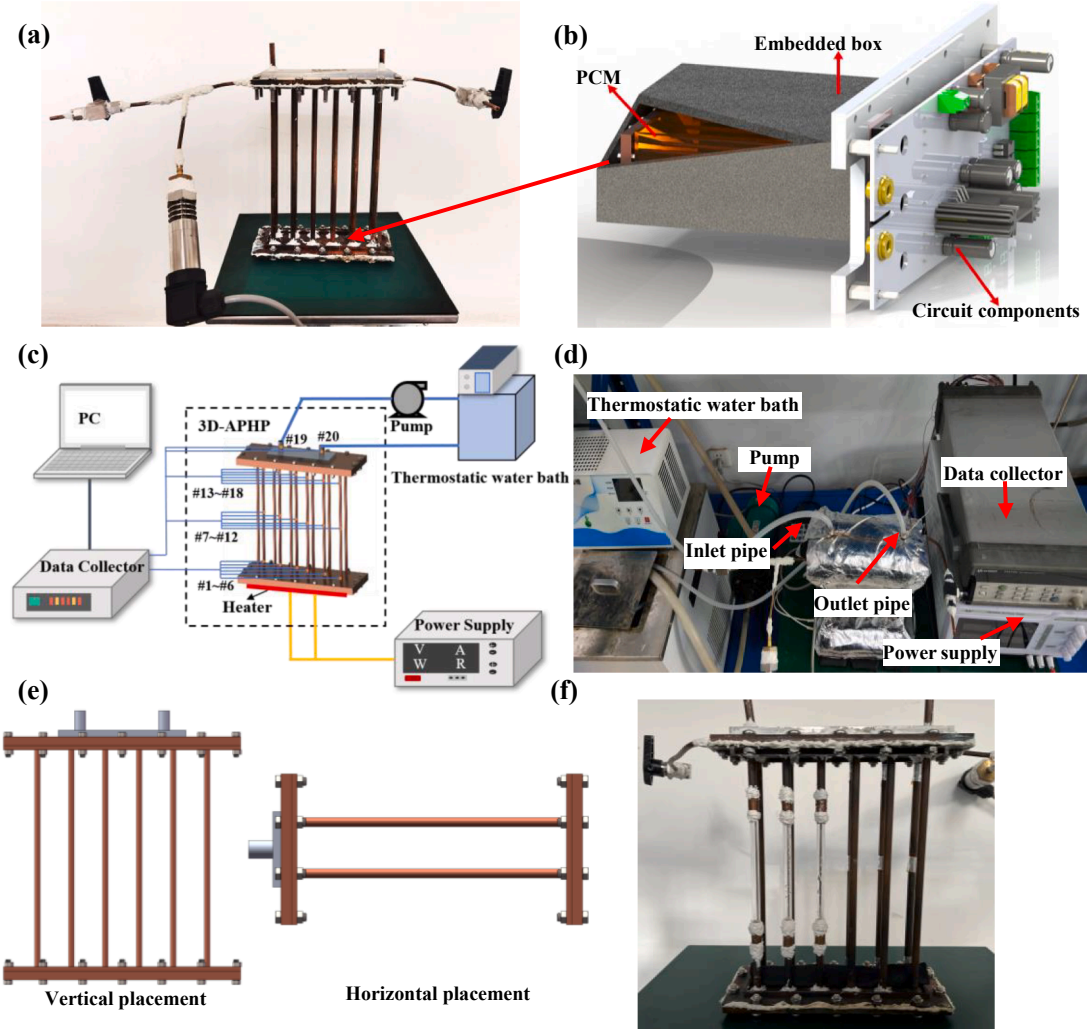


Fig. 2. (a) Physical schematic of 3D-APHP; (b) Schematic of insulated gate bipolar transistor module plug-in applying 3D-APHP and PCM coupling technology; (c) Schematic of 3D-APHP test system; (d) Physical schematic of 3D-APHP test system; (e) Schematic diagram of 3D-APHP under different installation conditions; (f) Visualization 3D-APHP.

Table 1
Thermophysical properties of working fluids.

Properties	Deionized water	Anhydrous ethanol	Novec7000
Liquid density [kg/m^3]	1000	789.34	1400
Liquid viscosity [μPa·s]	1010	1193.1	320
Liquid specific heat [kJ/kg· °C]	4.18	2.396	1.3
Latent heat [kJ/kg]	2260	926.01	142
Surface tension [mN/m]	72.8	22.35	12.4
Boiling point [°C]	100	78.4	34
Di_{max} [mm]	5.13	3.06	1.9

Table 2
Summary of experimental test conditions.

Adjustable parameters	Parameter Value
Working fluid	Deionized water, Anhydrous ethanol, Novec7000
Heating power [W]	20 ~ 100(Interval 10)
Filling rate [%]	5, 10, 15, 20, 30, 40, 50, 60, 70
Inner diameter [mm]	3, 4
Outer diameter [mm]	5, 6
Installation condition	Vertical, Horizontal

$$\Delta T_{ave} = \overline{T_e} - \overline{T_c} \tag{4}$$

$$R = \frac{\Delta T_{ave}}{Q} \tag{5}$$

The evaporation section to the condensation section of the PHP is viewed as a whole from a macroscopic perspective. By Fourier's law of thermal conductivity, the λ_e is calculated from formulas (6)~(7). The calculation of the λ_e is simplified by neglecting the R of the pipe wall due to the large thermal conductivity of purple copper [40].

$$Q = -\lambda A \frac{dT}{dx} \Rightarrow \lambda = \frac{Qdx}{A|dT|} \tag{6}$$

$$\lambda_e = \frac{4L_a}{\pi d_{a0}^2} \frac{Q}{\Delta T_{ave}} \tag{7}$$

In order to characterize the uniform temperature distribution of a 3D-APHP array, the η is defined as a characterization parameter of the uniform temperature characteristics, which is defined in formula (8). $\overline{T}$ is the overall average temperature of the 3D-APHP, and ΔT_{max} represents the maximum temperature difference [41]. A lower η indicates superior temperature uniformity in the 3D-APHP.

$$\eta = \frac{\Delta T_{max}}{\overline{T}} \tag{8}$$

Uncertainty analysis should be conducted prior to the experiment. In this experiment, directly measured physical quantities include volume, temperature, current, voltage and water velocity. Indirectly calculated quantities encompass FR, temperature difference, R, λ_e and η. Typically, errors in indirectly measured values originate from direct measurement inaccuracies. Based on statistical principles, the commonly used error transfer formula is a simplified form based on Taylor series expansion as shown in formula (9). Direct measurement precision is determined by equipment accuracy. Table 3 shows that the maximum error for indirect measurements is less than 5 %, indicating a minimal impact on experimental results.

$$\Delta y = \frac{\partial f}{\partial x1}\Delta x1 + \frac{\partial f}{\partial x2}\Delta x2 + \cdots + \frac{\partial f}{\partial xn}\Delta xn \tag{9}$$

3. Results and discussion

3.1. Thermal performance analysis of 3D-APHP under different working fluids

3.1.1. Wall temperature characterization

Fig. 3(a)~(c) corresponds to the wall temperature profiles of 3D-APHP with deionized water, anhydrous ethanol and Novec7000 as the working fluid, respectively. 3 mm is chosen as the inner diameter of 3D-APHP and the FR is selected as 30 %. During the heat storage process, the wall temperature of the 3D-APHP gradually increases. When the heating power is raised to 30 W, the wall temperature undergoes a sudden change, indicating that the working fluid has reached the saturation state and vaporization begins. When the heating power is further increased to 40 W, the operation of the PHP tends towards stability. As the heating power increases, the temperature difference between the evaporation section and the condensation section gradually decreases. Using deionized water as the working fluid results in significant temperature fluctuations during stable operation, with oscillation amplitudes reaching up to 20 °C. Switching to anhydrous ethanol dampens these wall temperature oscillations, notably reducing amplitude in both the adiabatic and condensation sections. With Novec7000 as the working fluid, wall temperature oscillations are largely suppressed.

The start-up and operation process of PHP are influenced by the combination of surface tension and viscosity of the working fluid. Lower surface tension shortens the bubble growth cycle, facilitating faster energy accumulation for boiling and quicker PHP start-up. According to Newton's equation of internal friction, it is known that the larger the viscosity, the larger the Newton's internal friction. Novec7000 ensures a rapid start-up process with its low viscosity and surface tension as shown in Table 1, whereas deionized water leads to a slower initiation due to its properties.

When the heat input is low, the temperature of the evaporation section and working fluid superheat are low, leading to weak phase transition intensity and small pressure pulsations from vaporization. This limits heat transfer drive, increasing the average temperature difference in the PHP. As heat input rises, gasification rate and pressure pulsation intensity surge, disrupting heat accumulation and enhancing the driving force. Consequently, the temperature in the condensation section increases, reducing the average temperature difference of the PHP.

Table 3
Summary of experimental uncertainties.

Designation	Accuracy	Unit
Heating power	2.5 %	W
Filling rate	2.5 %	%
Thermal resistance	4.5 %	°C/W
Thermal conductivity	4.5 %	KW/m °C

3.1.2. Comparison of heat transfer performance

The comparison of the R of 3D-APHP under different working fluids is shown in Fig. 3(d). On the whole, R shows a decreasing trend with the increased heat load. The decreasing trend of R becomes slower at high heat load, and R is less than 0.18 °C/W at the heating power of 100 W. Fig. 3(e) compares the λ_e of 3D-APHP. The variation trend of λ_e and R forms an opposite trend. The difference between R and λ_e of the three working fluids gradually decreases with the increasing heating power. In the pre-middle period of 3D-APHP operation (20 W ~ 60 W), the heat transfer performance is better when Novec7000 is used. When 3D-APHP is in the late period of operation (70 W ~ 100 W), the heat transfer performance is better when deionized water is employed, especially the R_{min} is 0.146 °C/W and λ_e maximum is close to 70 KW/m· °C when the heating power is at 100 W. Novec7000 exhibits the lowest latent heat of vaporization among the three working fluids, indicating that the evaporation section absorbs a smaller amount of heat to induce the evaporation and vaporization of the working fluid. Furthermore, the low viscosity and surface tension of Novec7000 facilitate faster fluid flow and reduce flow resistance during operation of PHP. As temperature rises, the impact of viscosity diminishes, while the significance of latent heat and specific heat capacity prevails. Deionized water exhibits high specific and latent heat values, which contribute to the enhanced intensity of the fluid phase transition. This leads to rapid bubble generation and intensified bubble perturbation in the liquid pool, ultimately augmenting nucleation boiling. Consequently, both the vaporization rate in the evaporation section and the liquefaction rate in the condensation section are elevated, creating a stronger driving pressure difference between the hot and cold ends. Based on these thermophysical properties, Novec7000 is more suitable for lower heat loads, while deionized water performs better under higher heat loads. When using three different working fluids, the R_{max} of the PHP does not exceed 0.98 °C/W, indicating that the 3D-APHP has the capability to adapt to a variety of working fluids.

3.1.3. Uniform temperature characteristic analysis

The array structure of the 3D-APHP is designed to ensure that the working fluid operates effectively in each copper tube, thus achieving overall uniform performance. Fig. 3(f) shows the comparison of η with different working fluids. When deionized water and anhydrous ethanol are used, η decreases rapidly firstly and slows down in the high-power region, and the overall uniform temperature characteristics gradually improve. While Novec7000 is used, η exhibits a decreasing trend, but the magnitude of change is not significant, and the uniform temperature characteristics show relatively minor alterations. The utilization of deionized water yields better overall temperature uniformity in the 3D-APHP with a minimum η of 0.252. Moreover, the overall η does not exceed 0.75. The uniform temperature performance is relatively poor when anhydrous ethanol and Novec7000 are used.

Due to the larger $Di_{\max}$ and latent heat of vaporization, deionized water exhibits outstanding heat carrying capability as the heat load increases. A large number of bubbles are generated at the vaporization core in the evaporation cavity. The vaporization rate rises rapidly to increase the pressure difference between the hot and cold ends, which accelerates the movement of the vapor plug in the tube. The PHP can form a stable vapor–liquid plug distribution between adjacent copper tubes, in the evaporation chamber and the condensation chamber, and the overall uniform temperature characteristics of 3D-APHP are excellent under these conditions. The small latent heat of vaporization of anhydrous ethanol and Novec7000 limits the rate and ability of heat transfer. Therefore, the uniform effect is poor. From the overall heat transfer process, the heat transfer performance of 3D-APHP with deionized water as the working fluid is relatively better.

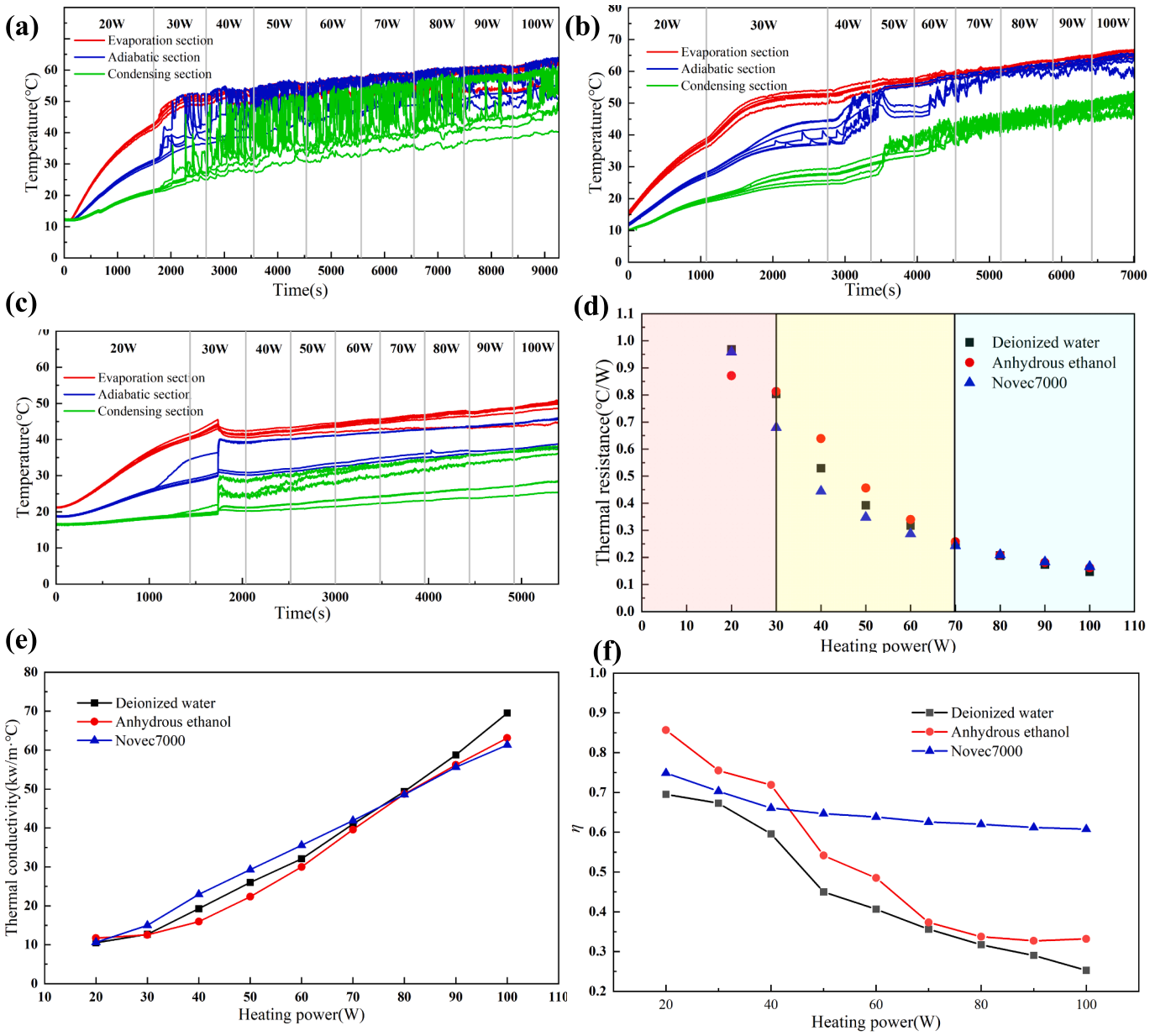


Fig. 3. Thermal performance analysis of 3D-APHP under different working fluids(Di = 3 mm, FR = 30 %):(a) Wall temperature distribution of deionized water-charged 3D-APHP ; (b) Wall temperature distribution of anhydrous ethanol-charged 3D-APHP ; (c) Wall temperature distribution of Novec7000-charged 3D-APHP ; (d) Comparison of thermal resistance of 3D-APHP ; (e) Comparison of thermal conductivity of 3D-APHP ; (f) Comparison of uniform temperature coefficient of 3D-APHP.

3.2. Thermal performance analysis of 3D-APHP at different FRs

3.2.1. Start-up performance analysis

Fig. 4(a) illustrates the start-up temperature of 3D-APHP at different FRs, ranging from 10 % to 70 %. 3 mm is chosen as the inner diameter and the deionized water is selected as the working fluid. It can be seen that the start-up temperature experiences a trend of first small increase, then rapid decrease and finally slow increase again. Notably, the lowest start-up temperature occurs at 20 % FR. Furthermore, it is observed that the average start-up temperature of deionized water is the highest, followed by anhydrous ethanol and finally Novec7000. The lower viscosity of anhydrous ethanol and Novec7000 helps maintain good fluidity at lower temperatures, facilitating the start-up of heat exchange processes. In addition, their lower boiling points mean that boiling and vaporization can begin at lower temperatures, resulting in lower start-up temperatures.

3.2.2. Comparison of heat transfer performance

The 3D schematic diagrams of the variation of R and λ_e of 3D-APHP at different liquid FRs are shown in Fig. 4(b) and (c). From the horizontal view, the R decreases gradually with the increase of heat input; from the vertical view, the R demonstrates a decreasing trend followed by an increasing trend as FR increases. A similar pattern is seen for λ_e, which initially rises then falls with increasing FR, but generally increases with heating power. Notably, when the heating power exceeds 70 W, the rate of increase in R slows and the decrease in λ_e becomes less pronounced. When the heating power is in the range of 20 ~ 50 W, the R drops steeply. As the heating power continues to increase, the PHP has enough heat input to drive the working fluid to full operation.

Comparing the R of 3D-APHP at different liquid FRs in the Fig. 4(b), it is evident that the heat transfer performance peaks at an FR of 20 % and a heat input of 100 W. The R and λ_e are 0.059 °C/W and 170.1 KW/m °C, respectively. These values greatly surpass the thermal

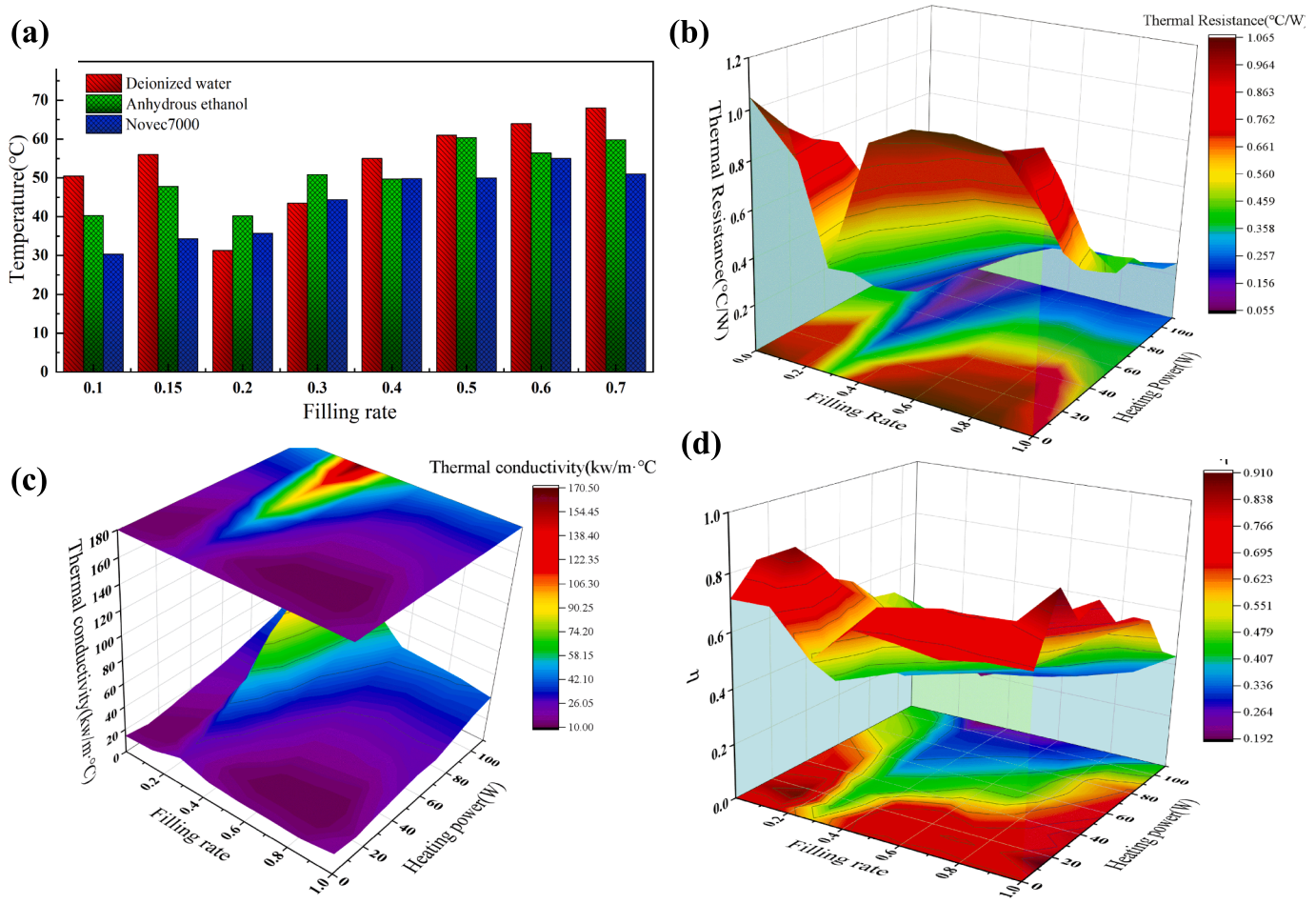


Fig. 4. Thermal performance analysis of 3D-APHP at different liquid FRs (Di = 3 mm, Deionized water): (a) Start-up temperature of 3D-APHP ; (b) Comparison of thermal resistance of 3D-APHP ; (c) Comparison of thermal conductivity of 3D-APHP ; (d) Comparison of uniform temperature coefficient of 3D-APHP.

conductivity of copper tubes, demonstrating the superior heat conduction performance of the 3D-APHP. Too low FR (10 %~15 %) and too high FR conditions (50 %~70 %) produce an adverse effect on the heat transfer performance of 3D-APHP. The smaller FR causes the liquid plugs to rise to a limited height, making it difficult to form a scouring effect on the condensation section, which leads to a low liquefaction rate and weak heat transfer. With the increase of FR, the mass and flow resistance of the working fluid also increase. Meanwhile, the liquid plug near the evaporation section is stronger in sensible heat absorption and heat transfer capacity, expanding the range of heat loads that the PHP can handle. The appropriate range of FR can reduce the phenomenon of liquid film fracture during liquid evaporation and condensation. When the FR is too large, it takes longer heating time at the same power to form bubbles at the vaporization core. This leads to a significant reduction in the pulsation frequency and speed of the formed vapor–liquid plug, resulting in deteriorated heat transfer. Compared with the small FR, the R of PHP is higher under the large FR. Based on the distribution state of the thermal performance curve of 3D-APHP, it can be presumed that the optimum FR is within the range of 20 %~40 %, allowing PHP to operate stably and exhibit better heat transfer performance. PHP is more suitable for high-power heating when FR is larger. The average R of PHP does not exceed 0.97 °C/W and the average λ_e is not less than 10 KW/m °C throughout the overall operation, it suggests that the 3D-APHP supports a wide range of FRs.

3.2.3. *Uniform temperature characteristic analysis*

Fig. 4(d) corresponds to the variation of the uniform temperature coefficient of 3D-APHP at different FRs. The η is similar to the trend of R. The overall distribution of η remains relatively consistent without excessively high or low values, and the average η does not exceed 0.9 during the operation. Notably, the best temperature uniformity performance of 3D-APHP is achieved at 20 % FR with a η_{min} of 0.193. The overall η does not exceed 0.85, indicating that the 3D-APHP possesses excellent temperature uniformity performance.

The temperature uniformity performance of the PHP is influenced by the total amount of working fluid and the state of slug motion. At lower FR, due to a small total volume of working fluid, it is speculated that the proportion of vapor slugs within the 3D-APHP exceeds that of liquid slugs. A stable pulsating flow cannot be formed inside the tube, leading to heat transfer obstruction and poor temperature uniformity in the 3D-APHP. When the FR is in the range of 20 %~50 %, the vapor–liquid plugs inside the PHP reach a dynamic equilibrium. It can ensure that there is enough thermal driving force to push the working fluid reciprocating movement, but not easy to occur the phenomenon of “dry-out”, making it the operational FR for the 3D-APHP. Once the FR exceeds 60 %, it’s assumed that the phenomenon of liquid slugs outnumbering vapor slugs emerges within the 3D-APHP. Part of the liquid in the heating section is directly inputted by the tube wall, while the liquid in the adiabatic section and condensation section relies on convection heat transfer and the temperature rises more slowly. At this time, the overall uniform temperature performance of PHP is poor. The reasonable FR can ensure the uniform temperature of the 3D-APHP.

3.3. Thermal performance analysis of 3D-APHP at different tube diameters

The size of the tube diameter has a very obvious impact on the heat transfer limit of PHP. Assuming that the tube diameter is smaller, it will increase the frictional resistance and pressure drop losses; while the tube diameter becomes larger, it will increase the thermal response time. In this section, the effect of tube diameter on the thermal performance index of 3D-APHP is discussed. The deionized water is chosen for the working fluid and the FR is selected as 20 % (optimal FR).

Fig. 5(a) compares the start-up temperature of 3D-APHP with different tube diameters. With the increase of FR, the start-up temperature of PHP has gone through the process of decreasing and then increasing, and the start-up temperature is the lowest at 20 % FR and the highest at 70 % FR. When $Di = 4$ mm, the start-up temperature of PHP is relatively low.

Fig. 5(b)–(d) corresponds to the variation trend of R, λ_e and η of 3D-APHP at different tube diameters, respectively. In the 20 W to 60 W power range, decreases in R and η and increases in λ_e are pronounced, but this trend moderates from 60 W to 100 W. When the Di is 3 mm, 3D-APHP demonstrates relatively superior thermal performance parameters with R_{min} at 0.059 °C/W, maximum λ_e at 120.1 KW/m °C, and η_{min} at 0.193. With Di at 4 mm, the thermal performance drops: R_{min} increases to 0.12 °C/W, λ_e decreases to 64.3 KW/m °C, and η_{min} rises to 0.31. This suggests that increasing the pipe diameter does not necessarily improve the heat transfer performance of PHP. Under the same heat load conditions, reducing the pipe diameter effectively enhances the unit heat flux of the 3D-APHP, leading to an increase in axial heat flux density. As a result, nucleate boiling is promoted and vaporization rate is improved. Moreover, a smaller pipe diameter implies a thinner liquid film, enhancing the heat exchange effect between the liquid and gas phases. Table 1 shows that the Di_{max} of deionized water is 5.13 mm. The Di_{max} will gradually decrease as the temperature rises, thus affecting the vapor–liquid distribution inside the tube. If the tube diameter is larger, the influence of the Di_{max} may inhibit the reflux of the working fluid. Consequently, a greater driving force is required to maintain the continuous oscillation in the tube.

The λ_e gradually increases and approaches its extreme value before the PHP reaches its heat transfer limit. With $Di = 3$ mm, the λ_e of PHP can be 1 to 2 orders of magnitude, corresponding to the unit of KW/(m•°C). This is an order of magnitude larger than the λ_e of PHP with $Di = 4$ mm, and λ_e can be more than 424 times larger than that of copper.

3.4. Thermal performance analysis of 3D-APHP under different installation conditions

Fig. 6(a) and (b) compares the start-up temperature under vertical and horizontal installation conditions. It can be clearly found that the start-up temperature at horizontal orientation is increased by 10 ~ 20 °C on average under different FRs when compared with vertical orientation, indicating that the start-up process of the PHP is more difficult in the horizontal direction. Due to the absence of gravitational assistance in a horizontal direction, the flow of vapor plugs within the PHP solely relies on the pressure difference generated between the hot and cold ends. This restricts the reciprocating motion of the working fluid and can obstruct the movement of the vapor plugs, thus necessitating a higher startup temperature.

Gravity plays a role in promoting the reflux of the working fluid in

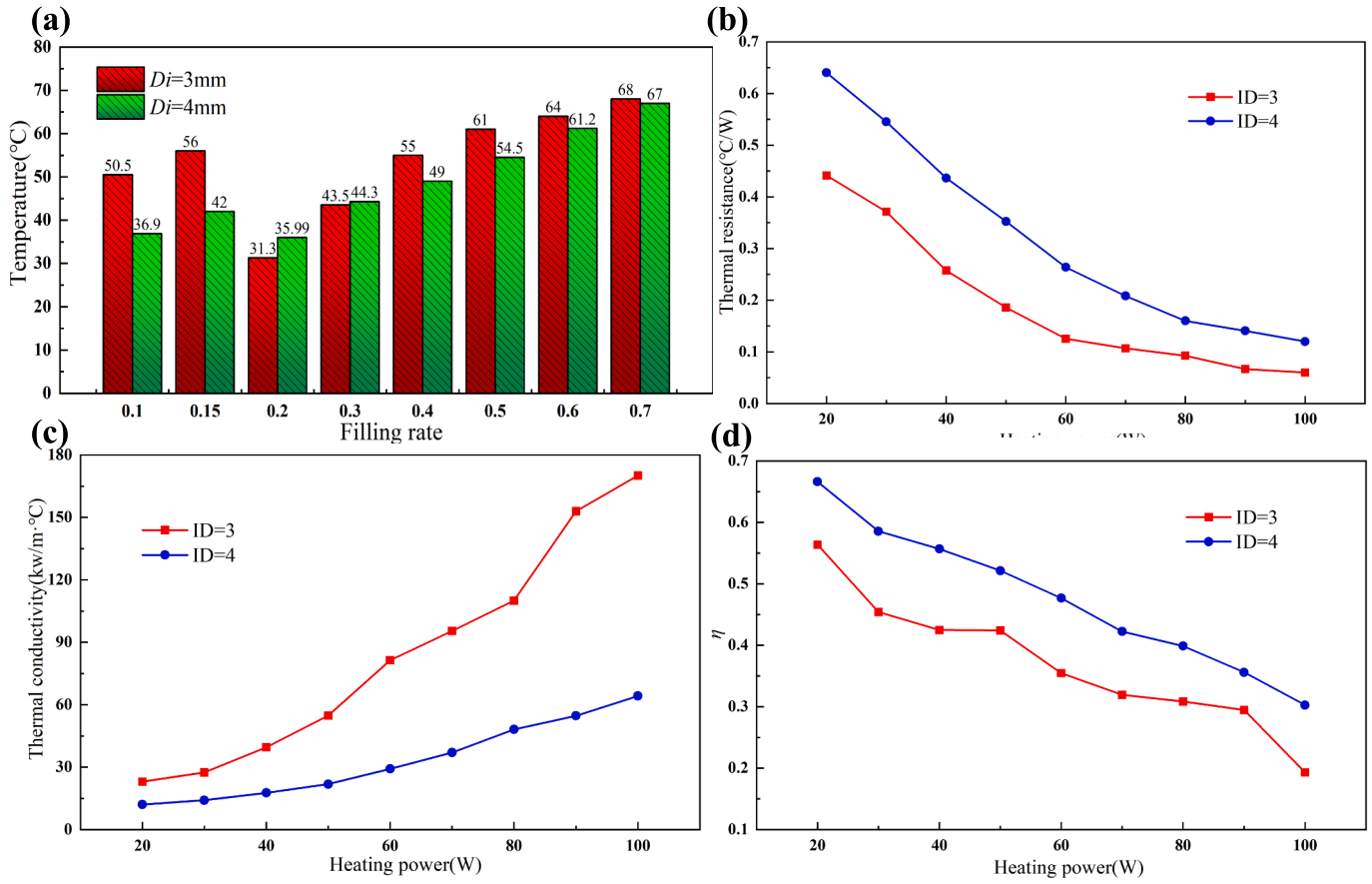


Fig. 5. Thermal performance analysis of 3D-APHP under different pipe diameter conditions (FR = 20 %, Deionized water): (a) Start-up temperature of 3D-APHP ; (b) Comparison of thermal resistance of 3D-APHP ; (c) Comparison of thermal conductivity of 3D-APHP ; (d) Comparison of uniform temperature coefficient of 3D-APHP.

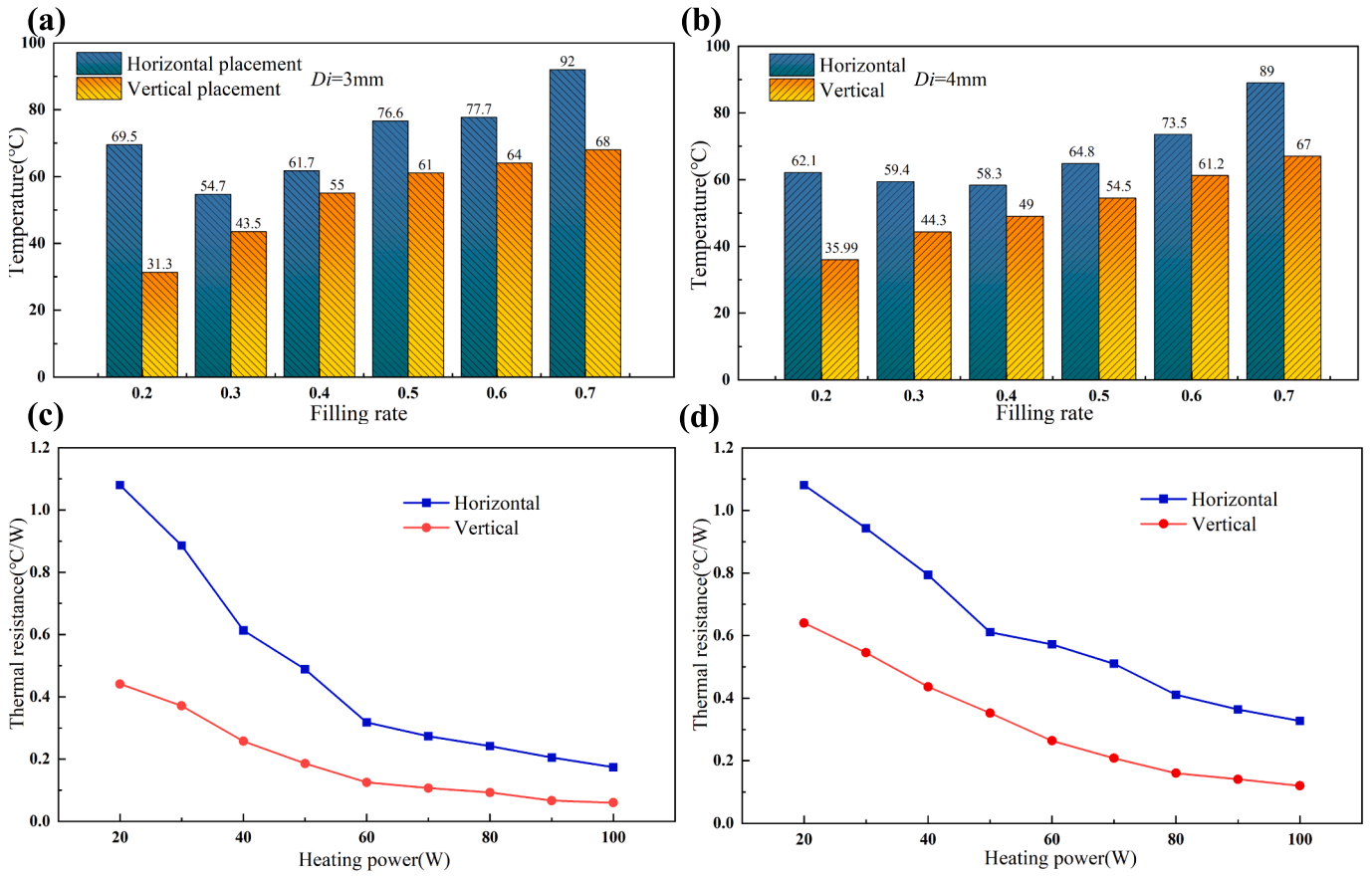


Fig. 6. Thermal performance analysis of 3D-APHP under different placement conditions (Deionized water): (a) Start-up temperature of 3D-APHP (*Di* = 3 mm); (a) Start-up temperature of 3D-APHP (*Di* = 4 mm); (c) Comparison of thermal resistance of 3D-APHP (FR = 30 %, *Di* = 3 mm); (d) Comparison of thermal resistance of 3D-APHP. (FR = 30 %, *Di* = 4 mm).

the bottom heating mode. But the effect of gravity in promoting the reflux is reduced when deviating from the vertical position, which leads to the reduction of heat transfer capacity and directly makes a difference to the reflux of the working fluid and the burn-dry limit of the PHP. The comparison of *R* for 3D-APHP under different installation conditions is presented in Fig. 6(c) and (d). 30 % FR is selected for this analysis. It can be observed that under the horizontal direction, the overall *R* of PHP for both tube diameter conditions is relatively high. Compared with the vertical direction, *R* increases by 50 % on average, and the R_{max} reaches 1.08 °C/W during the start-up of PHP. Since the working fluid is not completely concentrated in the evaporation section, so the heat source cannot achieve uniform heating of all the working fluid. The liquid reflux is not replenished in time, resulting in high temperature in the evaporation section and reduced heat transfer capacity. After comparing the conclusion with the above one, it is found that the overall heat transfer performance of 3D-APHP in vertical direction is better. It can start working at lower temperature with relatively low *R*. Nonetheless, the orientation of 3D-APHP in a horizontal installation configuration not only guarantees its regular and stable functionality but also fulfills the thermal management criteria. This adherence aligns with the initial design's "plug and play" strategy and is in line with the feature of ease of installation.

3.5. Analysis of the internal flow mechanism of 3D-APHP

The heat transfer process of 3D-APHP has both latent heat transfer from the vapor–liquid phase change and sensible heat transfer from the reflux process of the working fluid. In this section, the flow pattern and heat transfer mechanism inside the 3D-APHP with three working fluids are explored in depth.

Fig. 7(a)–(c) displays the infrared thermal images of the 3D-APHP under three different working fluid conditions, clearly demonstrating the flow state. The selected FR is 30 %. When the heating power is at 20 W, the evaporation section heats up rapidly without significant temperature rise in the condensation section, indicating that the PHP is not activated at this point and maintains a relatively low temperature. A comparison between different working fluids reveals that using Novec7000 enables faster heat transfer under low heat load conditions, with the lack of pulsating effect inside the PHP making its heat transfer performance similar to that of a steam chamber. In contrast, the temperature rise of water and ethanol is slower. As the heating power increases to 60 W, the 3D-APHP efficiently transfers heat to the condensation section. The uniform temperature distribution and clear gradients are observed across the tubes. When the heating power reaches 100 W, the flow process inside the PHP is more rapid and the temperature difference between neighboring copper tubes is obvious.

Fig. 8(a)–(d) shows the infrared thermograms of 3D-APHP under different working fluid conditions with a fixed heating power of 60 W. When FR is 10 %, the temperature difference between the evaporation section and the condensation section is significant and the bottom temperature is higher due to the low total amount of working fluid. With the increase of FR, the temperature of condensation section is obviously higher and the overall temperature distribution of the 3D-APHP becomes more uniform. When Novec7000 is used, there is a notable temperature difference between adjacent copper tubes. When deionized water is used, the PHP has more excellent temperature uniformity performance, which is accordance with the conclusions drawn from the above study.

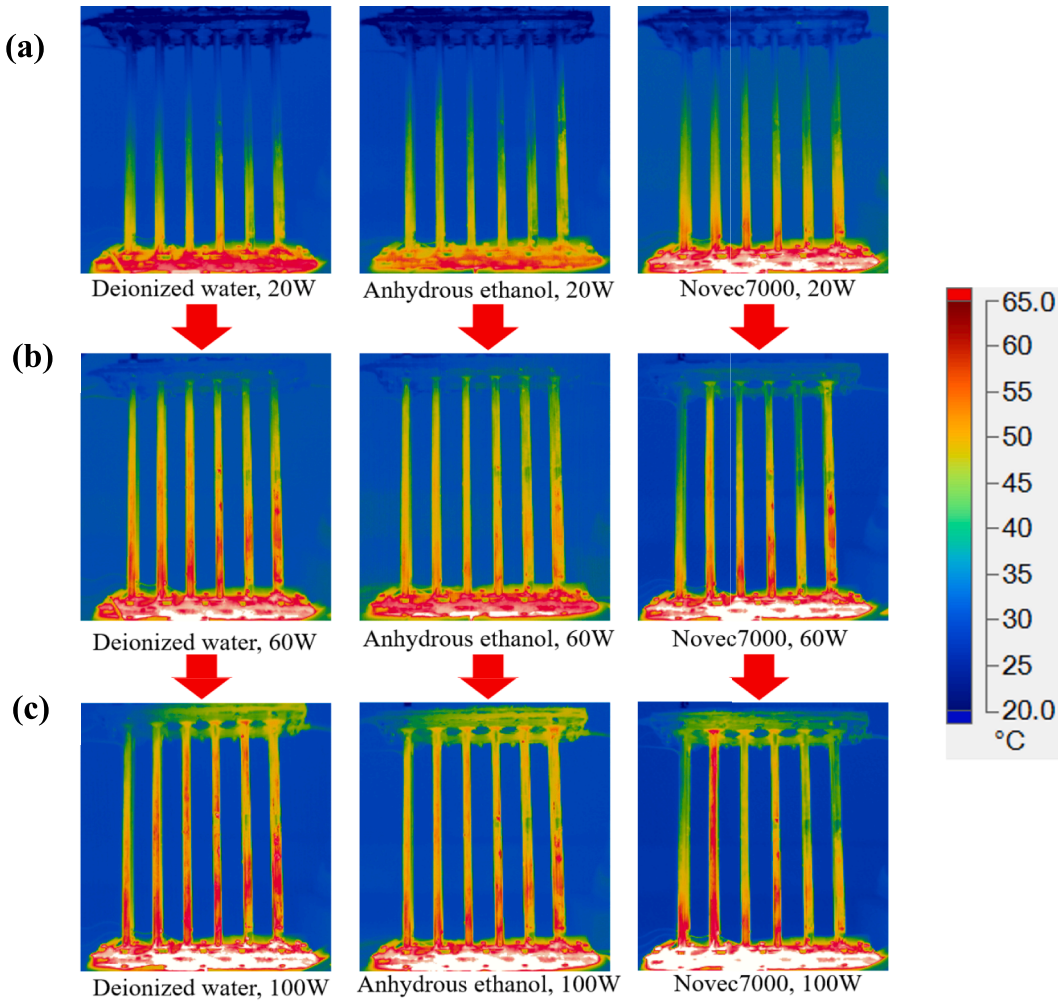


Fig. 7. Schematic of 3D-APHP thermal imaging under different working fluid and heating power conditions (FR = 30 %, *Di* = 3 mm): (a) Q = 20 W; (b) Q = 60 W; (c) Q = 100 W.

Fig. 9(a)–(c) speculates the flow state of the working fluids after the stable operation of the 3D-APHP and corroborated by visualization experiments. The two-phase heat transfer flow state varies significantly with different working fluid conditions. In this study, the inner diameter of 3D-APHP array tube as well as the height of evaporation and condensation chambers are 3 mm. As can be seen from Table 1, the Di_{max} corresponding to deionized water, anhydrous ethanol and Novec7000 to meet the alternating vapor–liquid plug distribution gradually decreases to 5.4 mm, 3.4 mm and 1.9 mm in order. As shown in Fig. 9(a), the 3D-APHP array tube, evaporation chamber and the condensation chamber form an obvious vapor–liquid plug distribution when deionized water is used. Vapor bubbles nucleate, grow rapidly and enlarge in the evaporation section, driving liquid movement. With the timely replenishment of the condensing back liquid and the pressure change brought by the vapor bubble rupture, the PHP will temporarily reach thermal equilibrium, waiting for the next work. In addition, the large surface tension of water can help maintain a clear liquid–gas interface. Combined with the above reasons, it causes a low internal liquid plug pulsation rate and leads to a high amplitude of wall temperature pulsation of the array tube. As shown in Fig. 9(b), when anhydrous ethanol is used, the Di_{max} gradually decreases to below 3 mm with the increase of temperature. The Di_{max} may be lower than 3 mm due to the high temperature in the evaporation section, and the stable alternating distribution of the vapor–liquid bridge (plug) cannot be formed in the evaporation section. The occurrence of temperature pulsation at the PHP wall is negligible. While the temperature of the adiabatic section and the condensation section is lower, the Di_{max} is higher than 3 mm and consequently the internal can form a stable alternating distribution of vapor–liquid bridge (plug). The frequency of the array tube wall temperature pulsation increases and the amplitude decreases. As shown in Fig. 9(c), the Di_{max} is much less than 3 mm when Novec7000 is used. The internal cannot be enough to form alternating distribution of vapor–liquid bridges (plugs) and a long liquid film is formed instead.

In the operation of PHP, the length and position of the vapor plug, the length and position of the liquid plug and the thickness of the liquid film are somewhat random. Both the length and speed of the vapor plug have a great influence on the ratio of heat transfer between the two regions of the vapor plug. When the length of the vapor plug is fixed, film evaporation is continuously generated as the velocity of the vapor plug increases, enhancing the film evaporation process. Consequently, the proportion of heat transfer through the liquid film also increases. The length of the liquid plug has an important influence on the heat transfer process. Under the same pressure difference, the liquid plug with smaller length can produce stronger oscillation and the oscillation amplitude will be larger. Therefore, the corresponding liquid plug heat transfer will be high. This explains why the internal oscillation phenomenon of PHP is more obvious when water is used as the working fluid. As the length of the liquid plug increases, the oscillation of the liquid plug decreases. The heat transfer capacity between the liquid plug and the wall also decreases, hence the heat transfer of the liquid plug

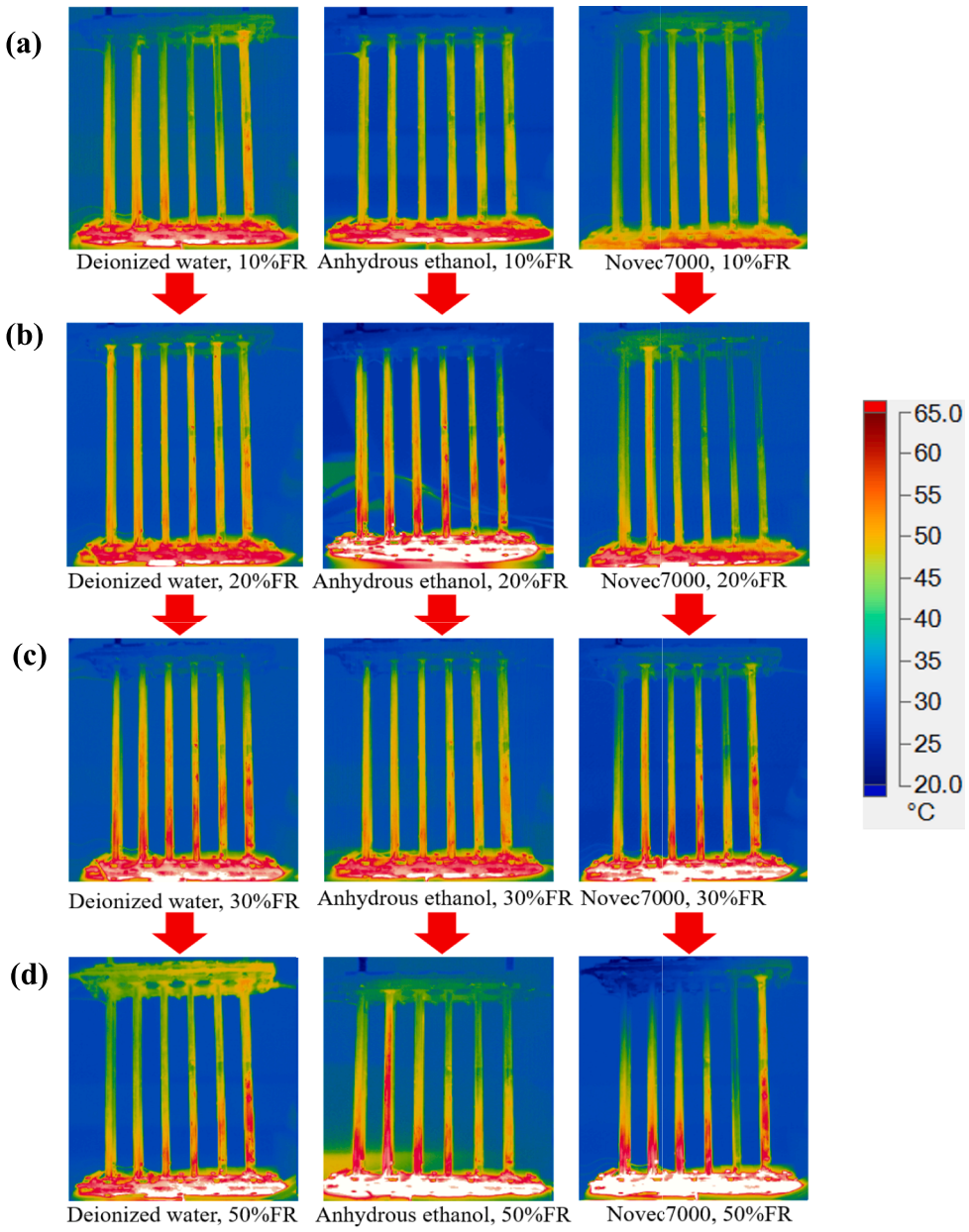


Fig. 8. Schematic of 3D-APHP thermal imaging under different FR and working fluid conditions (Q = 60 W, *Di* = 3 mm): (a) FR = 10 %; (b) FR = 20 %; (c)FR = 30 %; (d)FR = 50 %.

will gradually decrease. As the flow speed of the liquid plug increases, the total heat transfer between the liquid plug and the wall surface increases.

4. Conclusion

In this work, a 3D-APHP with a dual-plane structure was designed. The planar configuration facilitates contact with flat heat sources, reducing the contact thermal resistance and enhancing the convenience of installation. The array structure effectively improves temperature uniformity, broadening its range of applications. The 3D-APHP displays excellent heat transfer and temperature uniformity performance under various working conditions, making it suitable for the majority of electronic cooling applications. The main conclusions are as follows:

(1) At low heating power, the anhydrous ethanol and Novec7000 have advantageous effects on the low-temperature operation of 3D-APHP. For high heating power, deionized water with its higher latent heat value has the ability to carry a larger amount of heat. This results in better heat transfer capabilities and ensures the attainment of thermal equilibrium in the system.

(2) The operational filling rate range for the 3D-APHP lies between 20 % and 40 %. Both excessively low and high filling rates can affect the heat transfer performance of the 3D-APHP. The 3D-APHP demonstrates optimal overall heat transfer performance when the filling ratio is at 20 % and the heating power is at 100 W, with thermal resistance and thermal conductivity being 0.059 °C/W and 170.1 KW/m °C, respectively. The temperature uniformity performance is excellent with a uniformity coefficient value of 0.193.

(3) Under vertical installation conditions, the 3D-APHP with a tube diameter of 3 mm exhibits relatively superior operational characteristics and heat transfer capability. The equivalent thermal conductivity of the 3D-APHP can reach from 1 to 2 orders of

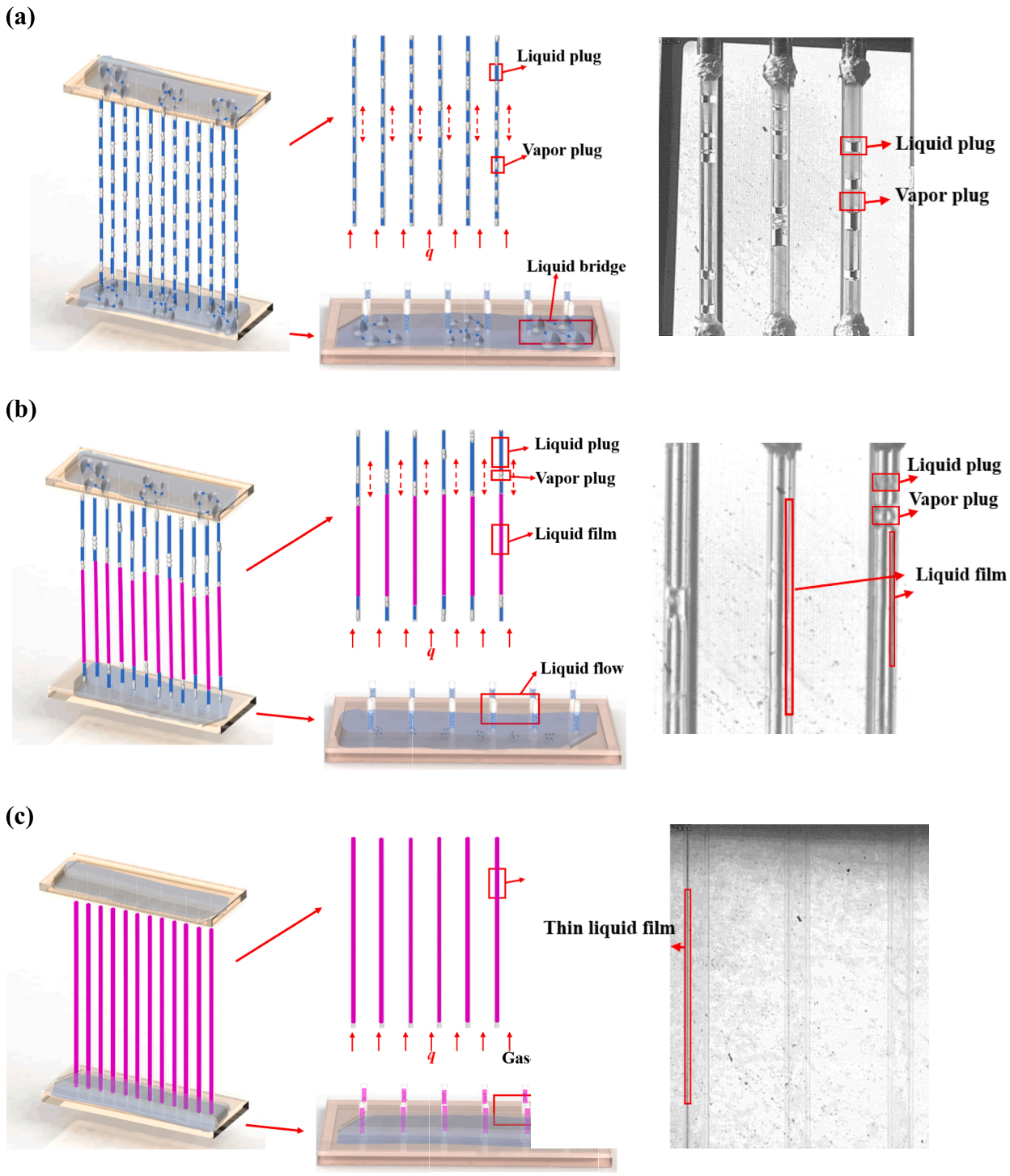


Fig. 9. Internal flow process of 3D-APHP under different working fluid conditions: (a) Deionized water-charged 3D-APHP; (b) Anhydrous ethanol-charged 3D-APHP; (c) Novec7000-charged 3D-APHP.

magnitude, expressed in KW/(m °C). Additionally, the designed 3D-APHP is capable of satisfying the heat dissipation needs in a horizontal direction, with an overall thermal resistance not exceeding 1.08 °C/W. This expands the range of applications and aligns with the "plug and play" policy.

(4) The heat transfer process in the 3D-APHP is influenced by the distribution of vapor–liquid plugs and the condition of the liquid film. The critical pipe diameters corresponding to deionized water, anhydrous ethanol, and Novec7000 gradually decrease to meet the alternating distribution requirements of vapor–liquid plugs. When using deionized water as the working fluid, the critical pipe diameter is at its maximum. In the 3D-APHP, the vapor–liquid plugs are distributed alternately and exhibit stable oscillating behavior.

CRediT authorship contribution statement

Yucheng Dai: Investigation, Methodology, Writing – original draft, Writing – review & editing, Data curation, Formal analysis. **Ren Zhang:** Investigation, Writing – review & editing. **Zhongyang Qin:** Formal

analysis. **Kaibao Liu:** Formal analysis. **Changhui Liu:** Validation. **Jiateng Zhao:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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